Duration Models: From Hazards to Treatment Effects

Motivated by Abbring & Van den Berg (2003, *Econometrica*)

Econometrics Reading Group

Why duration models?

- Many economic outcomes are about **how long until something happens**:
  - unemployment → job; poverty → exit; firm → closure; marriage → divorce.
- The questions are about the **rate** of leaving:
  - how it changes with time already spent;
  - how it responds to events along the way.

Destination of this talk: Abbring & Van den Berg (2003)

- The effect of a treatment that *arrives during the spell* (e.g. a benefits reduction) on the exit rate.
- To get there, we build the objects one at a time.

Road map: the questions, in order

1. **How do we describe a duration?**
   Hazard and survival. Cox (1972), Lancaster (1979)

2. **People differ: is a falling exit rate duration dependence or sorting?**
   Unobserved heterogeneity, dynamic selection. Lancaster (1979), Heckman & Singer (1984)

3. **When can the two be separated?**
   The mixed proportional hazard model. Elbers & Ridder (1982)

4. **Two durations racing.**
   Competing risks. Cox (1959), Tsiatis (1975), Heckman & Honoré (1989)

5. **A treatment arriving during the spell.**
   Abbring & Van den Berg (2003)

One duration: hazard and survival

$Y \geq 0$ continuous, $S(y) := \Pr(Y > y)$ survival, f density.

$$\theta(y) := \lim_{dy \downarrow 0} \frac{\Pr(y \leq Y < y + dy \mid Y \geq y)}{dy} = \frac{f(y)}{S(y)} \qquad \text{[conditional probability]}$$

$$\Theta(y) := \int_0^y \theta(u)\, du = \int_0^y -\frac{S'(u)}{S(u)}\, du = -\log S(y) \qquad [S' = -f;\ S(0) = 1]$$

$$\implies \quad S(y) = e^{-\Theta(y)}$$

- $\theta(y)$: exit rate *among those still unemployed*. The effect will act on θ.
- **Duration dependence**: how $\theta(y)$ changes with time already spent.
  - θ increasing: *positive* (the longer unemployed, the more likely to leave).
  - θ decreasing: *negative* (the longer unemployed, the less likely to leave).
  - θ constant: none. $S(y) = e^{-cy}$, memoryless.

People differ: individual vs. population hazard

Unobserved type $V \geq 0$, $V \sim G$. Assume

$$\theta(y \mid V) = \lambda(y)\, V, \qquad \lambda(y) = \theta(y \mid V = 1), \qquad S(y \mid V) = e^{-\Lambda(y)V}.$$

What we observe is the **population**:

$$\bar{S}(y) = \mathbb{E}\big[S(y \mid V)\big] = \mathbb{E}\big[e^{-\Lambda(y)V}\big] \qquad \text{[LIE]}$$

$$\bar{\theta}(y) = \frac{\bar{f}(y)}{\bar{S}(y)} = -\frac{\bar{S}'(y)}{\bar{S}(y)} = \lambda(y)\, \frac{\mathbb{E}\big[V e^{-\Lambda(y)V}\big]}{\mathbb{E}\big[e^{-\Lambda(y)V}\big]} \qquad \text{[differentiate under } \mathbb{E}\text{]}$$

- The ratio is an average of V, weighting each type by its survival $e^{-\Lambda(y)V}$.
- Next: it is the average type **among those still unemployed**.

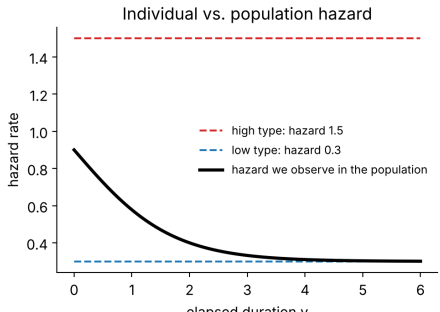
Who is still unemployed?

$$\Pr(V = v \mid Y > y) = \frac{S(y \mid v)\Pr(V = v)}{\bar{S}(y)} \quad \text{[Bayes]}$$

$$\mathbb{E}[V \mid Y > y] = \frac{\mathbb{E}[Ve^{-\Lambda(y)V}]}{\mathbb{E}[e^{-\Lambda(y)V}]} \quad \text{[its mean]}$$

$$\implies \bar{\theta}(y) = \lambda(y)\,\mathbb{E}[V \mid Y > y] \quad \text{[previous slide]}$$

- Prior $\Pr(V = v) \times$ likelihood $S(y \mid v)$: staying is evidence of a low v.
- Average individual hazard: $\lambda(y)\mathbb{E}[V]$.
- Observed hazard: $\lambda(y)\mathbb{E}[V \mid Y > y]$.
- Gap = **dynamic selection**.



Two types, constant hazards. Nobody's hazard falls; the observed one does.

The Laplace transform: isolating the type distribution

$$\mathcal{L}(a) := \mathbb{E}\big[e^{-aV}\big], \qquad \bar{S}(y) = \mathcal{L}\big(\Lambda(y)\big).$$

- $\mathcal{L}$ depends only on G; Λ only on duration dependence.
- Survival = (heterogeneity) applied to (duration dependence).

Properties

$$-\frac{\mathcal{L}'(a)}{\mathcal{L}(a)} = \mathbb{E}[V \mid \text{survived to } a] \qquad \text{[differentiate under } \mathbb{E}\text{]}$$

$$\frac{d^2}{da^2}\log \mathcal{L}(a) = \mathrm{Var}(V \mid \text{survived to } a) \qquad \text{[quotient rule]}$$

$$\implies \frac{d}{dy}\mathbb{E}[V \mid Y > y] = -\lambda(y)\,\mathrm{Var}(V \mid Y > y) \;\le 0 \qquad \text{[chain rule]}$$

- **Uniqueness:** $\mathcal{L}$ determines G.
- **Analyticity:** $\mathcal{L}$ on any small interval $\Rightarrow$ $\mathcal{L}$ everywhere.
- Strategy: learn the index, learn $\mathcal{L}$ on an interval $\Rightarrow$ G.

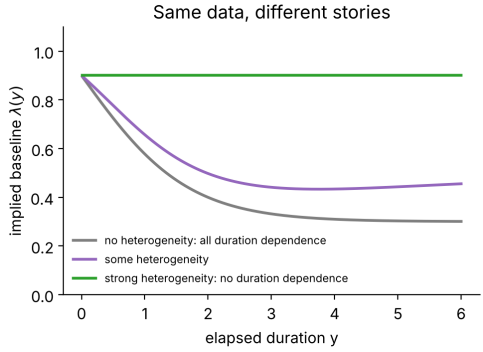
Duration dependence or sorting? Not identified from one duration

$$\bar{\theta}(y) = \underbrace{\lambda(y)}_{\text{duration dep.}} \cdot \underbrace{\mathbb{E}[V \mid Y > y]}_{\text{sorting}}$$

- Data: $\bar{S}(y)$ for all y.
- Unknowns: Λ and G, with $\bar{S}(y) = \mathcal{L}_G(\Lambda(y))$.
- One data function, two unknown functions of the same y.
- Any G fits: Λ bends to absorb the rest.

To separate them, we need variation that moves one but not the other:

- groups at different stages of sorting, same λ;
- the same person observed twice;
- parametric restrictions on Λ or G.



Same observed survival, three type distributions, three implied baselines.

Covariates separate sorting from duration dependence

MPH model: $\theta(y \mid x, V) = \lambda(y)\,\varphi(x)\,V, \quad V \perp X$. Data: $S(y \mid x)$ for all (y, x).

$$S(y \mid x) = \mathcal{L}\big(\varphi(x)\Lambda(y)\big): \quad \text{one } \mathcal{L} \text{ and one } \Lambda \text{ must fit } \textit{all} \text{ groups.}$$

Let $m(a) := -\mathcal{L}'(a)/\mathcal{L}(a) = \mathbb{E}[V \mid \text{survived to } a]$, decreasing. Compare $\varphi_1 > \varphi_2$:

$$\frac{\theta(y \mid x_1)}{\theta(y \mid x_2)} = \frac{\varphi_1}{\varphi_2} \cdot \frac{m(\varphi_1\Lambda(y))}{m(\varphi_2\Lambda(y))} \qquad [\lambda(y) \text{ cancels}]$$

$$< \frac{\varphi_1}{\varphi_2} \;\; (y > 0), \qquad = \frac{\varphi_1}{\varphi_2} \;\; (y = 0) \qquad [\varphi_1\Lambda > \varphi_2\Lambda,\ m \text{ decreasing}]$$

- At the same duration, the high-φ group is further along: its sorting is more advanced.
- Duration dependence hits both groups equally and cancels. Sorting does not.
- So the **drift of the ratio below φ_1/φ_2 measures sorting**.
- MPH reads: *absent sorting, covariate effects would be constant over duration*.

MPH identification in three steps

Elbers & Ridder (1982)

Normalize $\varphi(x^*) = 1$, $\Lambda(y^*) = 1$. Assume $\mathbb{E}V < \infty$ and variation in $\varphi(x)$.

1. $\dfrac{\theta(y \mid x)}{\theta(y \mid x^*)} \xrightarrow{y \downarrow 0} \varphi(x)$ [nobody sorted yet: $m(0)/m(0) = 1$]

2. $S(y^* \mid x) = \mathcal{L}\big(\varphi(x)\big)$ [$\Lambda(y^*) = 1$]

3. $\Lambda(y) = \mathcal{L}^{-1}\big(S(y \mid x)\big)/\varphi(x)$ [$\mathcal{L}$ strictly decreasing]

- Step 2: varying x traces $\mathcal{L}$ on an interval $\Rightarrow$ (analyticity) everywhere $\Rightarrow$ (uniqueness) G.
- One duration with covariates reveals λ, φ and G.
- Price: proportionality, $V \perp X$, $\mathbb{E}V < \infty$; identification leans on short durations.

A second duration: the treatment

Where it comes from

- A treatment that *arrives* at a random time S, with its own hazard, is itself a duration.
- Before S: a race between Y (job) and S (treatment), **competing risks**.
- After S: Y continues under treatment. The effect shows up here.

Why it is a hurdle

- Censoring: if the job comes first, S is never seen.
- Endogeneity: the same unobserved traits may drive both.
- Tsiatis (1975): race data alone cannot reveal that dependence.

Why modelling it is the solution

- If S were exogenous, it would be an ordinary time-varying covariate.
- Otherwise, model S's hazard: a dynamic **selection equation**.
  - Before S there is no effect yet: a clean window to learn the selection.

The race: model and data

$$\theta_Y(t \mid x, V) = \lambda_Y(t)\varphi_Y(x)V_Y, \qquad \theta_S(t \mid x, V) = \lambda_S(t)\varphi_S(x)V_S, \qquad (V_Y, V_S) \sim G.$$

Assumption: $Y \perp S \mid (x, V)$. All dependence runs through (x, V).

$$\begin{aligned} \Pr(Y > y, S > s \mid x) &= \mathbb{E}\big[e^{-\varphi_Y \Lambda_Y(y) V_Y} e^{-\varphi_S \Lambda_S(s) V_S}\big] && \text{[LIE; assumption]} \\ &= \mathcal{L}\big(\varphi_Y(x)\Lambda_Y(y),\ \varphi_S(x)\Lambda_S(s)\big) && [\mathcal{L}(a,b) := \mathbb{E}[e^{-aV_Y - bV_S}]] \end{aligned}$$

Only the first event is seen. With $a_t = \varphi_Y \Lambda_Y(t)$, $b_t = \varphi_S \Lambda_S(t)$:

$$P_0(t \mid x) := \Pr(\text{nothing yet by } t \mid x) = \mathcal{L}(a_t, b_t)$$
$$h_Y(t \mid x) := \text{exit rate among those with nothing yet} = \lambda_Y(t)\varphi_Y(x)\,\mathbb{E}[V_Y \mid R_t, x]$$

- R_t: the risk set, still unemployed and untreated at t.

Where the selection lives

Selection = dependence of V_Y and V_S among those at risk:

$$\frac{\partial^2 \log P_0}{\partial a \, \partial b} = \mathrm{Cov}(V_Y, V_S \mid R_t, x) \qquad \text{[quotient rule]}$$

$$\frac{\partial \log h_Y}{\partial b} = -\frac{\mathrm{Cov}(V_Y, V_S \mid R_t, x)}{\mathbb{E}[V_Y \mid R_t, x]} \qquad \text{[hold } a \text{ fixed]}$$

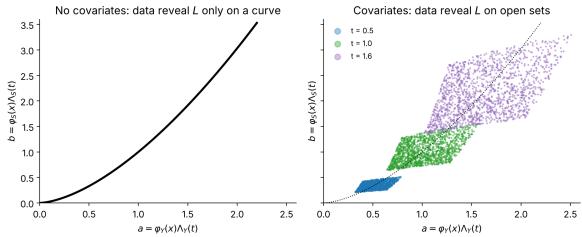
- Raise treatment intensity, hold the exit index fixed. Those at risk are untreated, so no causal channel.
- Any response of their exit rate is **pure selection**.

Assumption 3: $\{(\varphi_Y(x), \varphi_S(x))\}$ contains an open set of $\mathbb{R}^2$.

- E.g. $\varphi_j = e^{x'\beta_j}$, two covariates: direction $d = (\beta_{Y2}, -\beta_{Y1})$ gives $d'\beta_Y = 0$, $d'\beta_S \neq 0$ iff $(\beta_Y \ \beta_S)$ full rank.
- **vs. exclusion restriction:** no variable is excluded; a *direction* is, and it is learned from data.

Tsiatis' problem and how covariates solve it

- The surface $\mathcal{L}(a, b) = \Pr(\text{nothing yet})$ at indices (a, b). Selection lives in its shape.
- Each (group x, duration t) gives its height $P_0(t \mid x)$ at one location.
- **No covariates:** locations lie on one curve $t \mapsto (\Lambda_Y(t), \Lambda_S(t))$. Cannot pin down a surface.
- **Covariates (Assn. 3):** locations fill a 2-D patch $\Rightarrow$ surface $\Rightarrow$ G.



Three steps, as before:

1. h_Y, h_S ratios as $t \downarrow 0 \Rightarrow \varphi_Y, \varphi_S$
2. $P_0(t^* \mid x) = \mathcal{L}(\varphi_Y(x), \varphi_S(x)) \Rightarrow G$
3. solve for $\Lambda_Y(t), \Lambda_S(t)$

Adding the treatment effect

Abbring & Van den Berg (2003)

- Potential outcomes indexed by treatment date: $Y^*(s)$, $s \in [0, \infty]$. Observed $Y = Y^*(S)$.
- **No anticipation:** $\theta_{Y^*(s)}(t) = \theta_{Y^*(\infty)}(t)$ for $t < s$. The future date does not affect today's hazard.
- **Selection on** (x, V)**:** $Y^*(\cdot) \perp S \mid (x, V)$.

Model 1a

$$\theta_Y(t \mid s, x, V) = \lambda_Y(t)\varphi_Y(x)V_Y \cdot \delta(t \mid s, x)^{\mathbf{1}\{t>s\}}, \qquad \theta_S(t \mid x, V) = \lambda_S(t)\varphi_S(x)V_S.$$

- δ: the causal effect, a multiplier on the exit rate after treatment. May vary with t, s, x.
- Before S: exactly the race. So φ's, Λ's and G are identified.
- After S: the treated cohort's survival $P_1(y \mid s) = \Pr(Y > y \mid S = s, Y > s)$ carries δ.
- Given (x, V), the timing of S is still pure luck: the instrument-like variation.

Who is treated at s? The selection factor

$$dG_{R_s}(v) \propto e^{-a_s v_Y - b_s v_S}\, dG(v) \qquad \text{[Bayes: survivors at } s\text{]}$$

$$dG_s(v) \propto v_S\, dG_{R_s}(v) \qquad \text{[treated in } [s, s+ds) \text{ w.p. } \propto v_S\text{]}$$

Exit rate right after s, treated vs. not-yet-treated:

$$\frac{\text{just treated}}{\text{not yet treated}} = \delta(s \mid s, x) \cdot \frac{\mathbb{E}_{G_s}[V_Y]}{\mathbb{E}_{R_s}[V_Y]} \qquad [\lambda_Y \varphi_Y \text{ cancel}]$$

$$= \delta(s \mid s, x) \cdot \left(1 + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s} V_Y \; \mathbb{E}_{R_s} V_S}\right) \qquad [\text{reweight by } v_S]$$

- Observed ratio = causal effect $\times$ **selection factor**.
- No dependence $\Rightarrow$ factor 1: timing as good as random.
- The factor is a function of $\mathcal{L}$, known from the race $\Rightarrow$ δ identified.

After the moment of treatment

With $D(y \mid s, x) := \int_s^y \lambda_Y(\tau)\delta(\tau \mid s, x)\, d\tau$:

$$P_1(y \mid s, x) = \mathbb{E}_{G_s}\left[e^{-\varphi_Y(x)V_Y D(y|s,x)}\right] \qquad \text{[LIE over the treated cohort]}$$

$$= \frac{\mathcal{L}_b(a_s + \varphi_Y(x)D,\ b_s)}{\mathcal{L}_b(a_s, b_s)} \qquad \text{[the paper's eq. (10)]}$$

- Left: data. Right: known except D. Decreasing in $D \Rightarrow$ invert $\Rightarrow \delta = \partial_y D/\lambda_Y$.
- Two selection problems, both handled by knowing G:
  1. **into** treatment at s: the v_S tilt;
  2. **out of** the treated cohort after s: its survivors drift to low V_Y, even with random timing.

pre-treatment race $\Rightarrow \varphi, \Lambda, G \Rightarrow$ who is treated at $s \Rightarrow + P_1 \Rightarrow \delta$

What is traded

Not needed

- an excluded instrument
- selection on observables only
- parametric forms

Needed

- no anticipation (untestable)
- time-invariant unobservables, $V \perp X$
- proportional hazards (MPH)
- non-proportional covariate variation
- finite means; continuous time

The idea in one line

- The treatment acts only *after* S; unobserved types act throughout.
- The excluded *variable* becomes an excluded timing *shock*.

Caveats

- Identification leans on short durations.
- Discrete data blur the timing.
- δ is a hazard ratio given V, not a population effect.

Connections

Staggered-adoption DiD

- $Y^*(s) \leftrightarrow Y_t(g)$; the same no-anticipation assumption; risk set $\leftrightarrow$ not-yet-treated.
- Difference: an absorbing exit changes who remains, so parallel trends in hazards fails even with random timing.

Sequential exogeneity and feedback

- A hazard is an intensity given the past: a continuous-time sequential moment condition.
- Single spell: treatment may depend on permanent V, not on transitory shocks.

State dependence vs. permanent heterogeneity

- λ (duration dependence), G (heterogeneity), δ (intervention): the framework separates all three.
- Multiple spells identify more, but assume no causal links across spells.

Backup: any G fits one survival curve

Data $\bar{S}$. Pick any $\tilde{G}$ with $\Pr(\tilde{V} = 0) = 0$; we need $\mathcal{L}_{\tilde{G}}(\tilde{\Lambda}(y)) = \bar{S}(y)$ for all y.

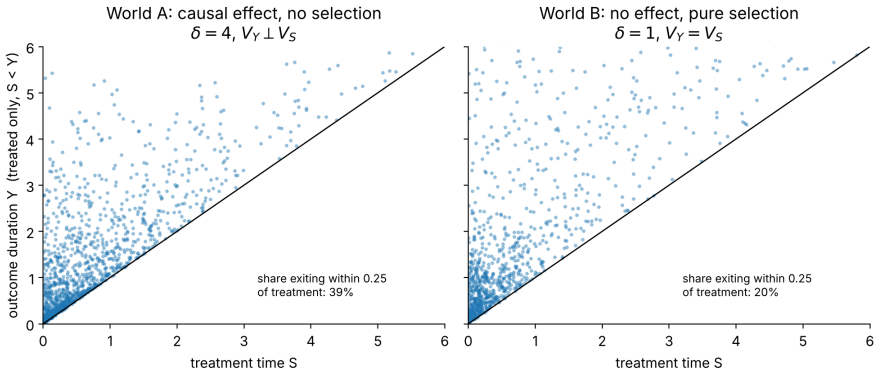
$$\tilde{\Lambda}(y) = \mathcal{L}_{\tilde{G}}^{-1}\big(\bar{S}(y)\big) \qquad [\mathcal{L}_{\tilde{G}} : [0,\infty) \to (0,1] \text{ strictly decreasing, onto}]$$

$$\tilde{\Lambda}(0) = 0, \quad \tilde{\Lambda} \text{ increasing} \qquad [\bar{S}(0) = 1\text{; composition of decreasing maps}]$$

$$\bar{\theta}(y) = \tilde{\lambda}(y)\,\tilde{\mathbb{E}}[V \mid Y > y] \qquad [\text{differentiate} - \log \text{ of the fit}]$$

- One equation in one unknown number $\tilde{\Lambda}(y)$ at each y: always solvable.

Backup: local vs. global dependence



Left: causal effect, no selection; exits bunch right after treatment at every S. Right: no effect, pure selection; S and Y correlated overall.